AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

T103/TM103
Computer Organization and Architecture

Midterm Examination (MTA)-Makeup
Spring Semester 2024/2025

Date: 24 /Apr./2025

lumber of Exam Pages: 2 ncluding this cover sheet)	**Time Allowed:** 90 mins

nstructions:

- Total Marks: **60**
- This exam consists of **TWO** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Programmable Calculators** are **NOT** allowed.

AOU TEST

CamScanner

Part 1: Multiple Choices questions [10 Marks]

Q-1: Answer all the following questions by choosing the most correct statement.

1- Which of the following is the lowest number system in which 584 could be a valid number?
   a. Decimal
   b. Octal
   c. Hexadecimal
   d. All the choices.

2- In the JK-Flip Flop, if K=1, J=0 then:
   a. Q(t+1)=1
   b. Q(t+1)= Q(t)
   c. Q(t+1)=0
   d. Q(t+1) is unknown.

3- 100 milliseconds are equivalent to:
   a. $100*10^{-3}$ microseconds
   b. $1*10^{-5}$ microseconds
   c. $10*10^{4}$ microseconds
   d. Both a and b

4- Which of the following is a valid statement?
   a. A word size can be 8 bits
   b. Full Adders have 3 inputs.
   c. D-F.F cannot be used in sequential circuits.
   d. Both a and b

5- What is the complement of the expression z'+ xy'?
   a. z. x'y
   b. z. (x'+y)
   c. z' + (x'y).
   d. None of the choices

PART 2: ALL QUESTIONS ARE REQUIRED [50 Marks]

Q-2: Below are the equations of the Boolean function G:

G=x(y'+z)

Draw the Truth Table of G showing the intermediate results (y' & y'+z). **[5 marks]**

Q-3: Given the floating-point notation excess16 diagram. Find the decimal number that it represents. **[7 marks]**

FP-excess 16



Q-4: Perform the following base conversions: **[12 marks]**

a) $(118)_{10}$ = ( )$_2$ [2 marks]

b) $(10101110)_2$ = ( )$_8$ [2 marks]

c) $(37.4)_8$ = ( )$_{10}$ [4 marks]

d) $(EA)_{16}$ = ( )$_8$ [4 marks]

Q-5 : Represent the decimal number (-91) in binary using 8-bit sign magnitude, One's complement, and two's complement: **[12 marks]**

Q-6 : Given the SOP of F: **F= x'y'z + x'yz' + xyz +x'yz + xyz'.** **[14 marks]**

a. Draw the K-map of F, showing the groups that you have selected. (6 marks)
b. Deduce, from part a, the simplified F. (4 marks)
c. Draw the logical Diagram of the simplified F. (4 marks)

End of questions

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